

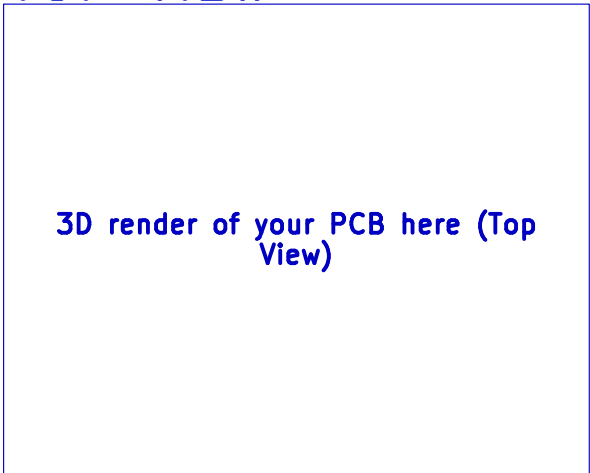
NU0001_DRS – Controller

VARIANT: DRAFT

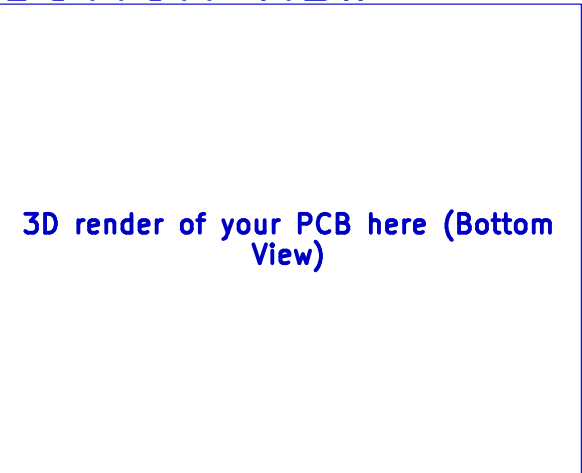
2025-09-16
REVISION 0

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TOP VIEW



BOTTOM VIEW



NOTES

Add a comment here
Not fitted components are marked as **X**
DRAFT – Very early stage of schematic, ignore details.
PRELIMINARY – Close to final schematic.
CHECKED – There shouldn't be any mistakes. Contact the engineer otherwise.
RELEASED – A board with this schematic has been sent to production.

DRAFT Release Date

DESIGN CONSIDERATIONS

DESIGN NOTE:
Example text for informational design notes.

DESIGN NOTE:
Example text for debug notes.

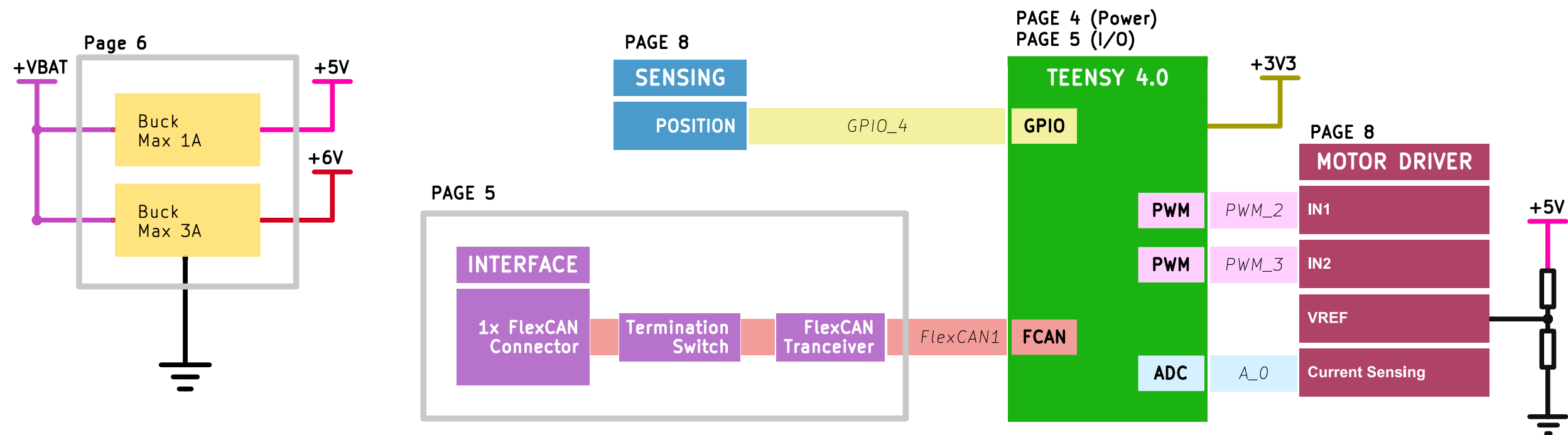
DESIGN NOTE:
Example text for cautionary design notes.

DESIGN NOTE:
Example text for critical design notes.

LAYOUT NOTE:
Example text for critical layout guidelines.

NOTES		 		NU Teams THIS DESIGN AND/OR DRAWING SHALL NOT BE REPRODUCED WITHOUT AUTHORISATION OF THE DESIGNER AND/OR NU TEAMS.	
		TITLE Root		CREATED DATE 2025-09-12	
DSG NR.	R. HICKS	PART NO	DRAWING NO NU0001-ELEC-GA-701		SHEET SIZE A3
ENG.					
CHKR.	Reviewer Name				
APPR.	Approver Name	KiCad E.D.A. 9.0.3	REVISION 0	SHEET 1 OF 10	

[2] Block Diagram



TARGET SPECIFICATIONS

INPUT VOLTAGE	12 – 14V
VOLTAGE SPEC #	??
VOLTAGE SPEC #	??
VOLTAGE SPEC #	??

NOTES



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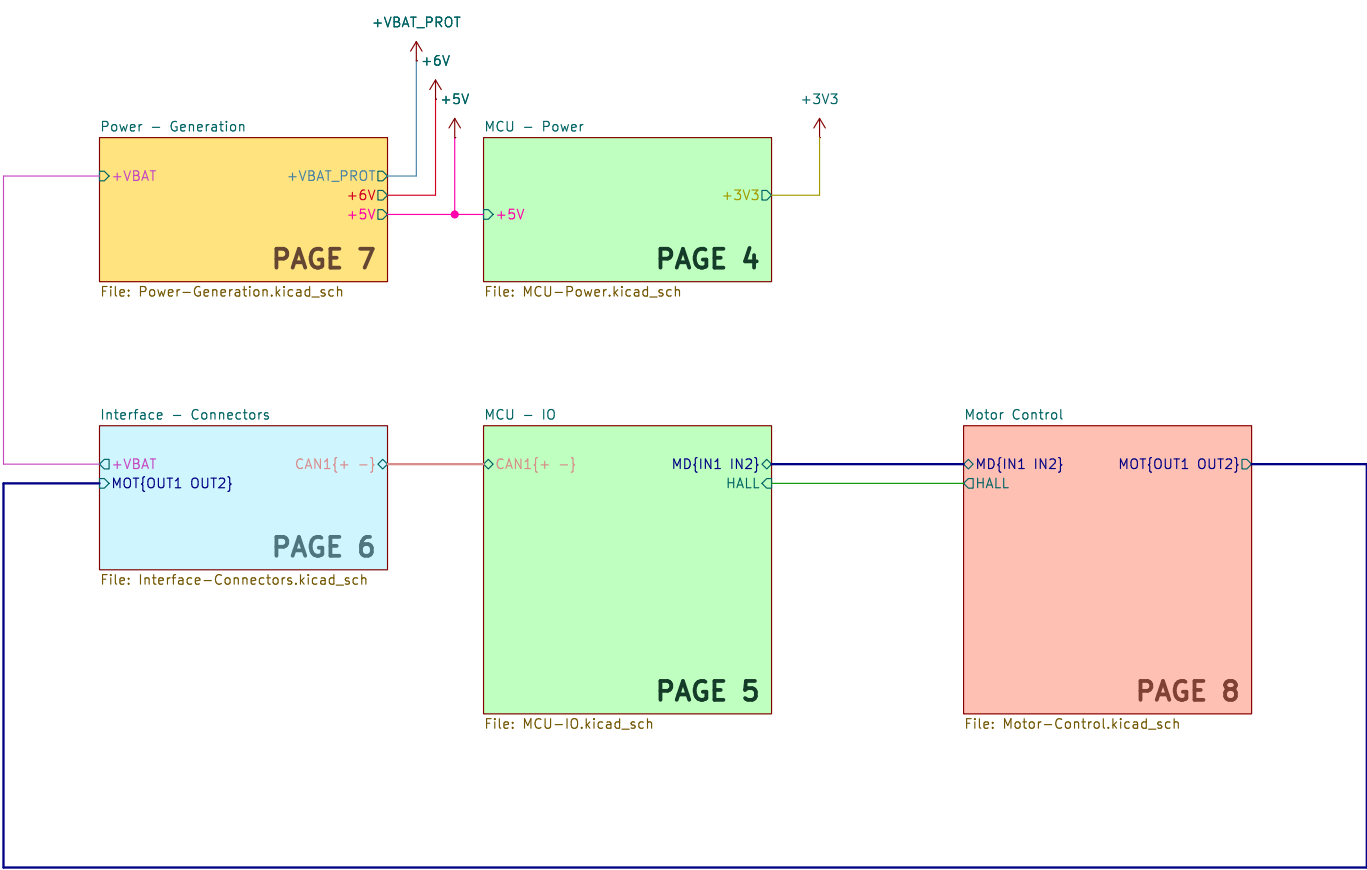
TITLE
Block Diagram

CREATED DATE 2025-09-12

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ENG.	
CHK R.	Reviewer Name
APPR.	Approver Name

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	NU0001-ELEC-BD-701	A3
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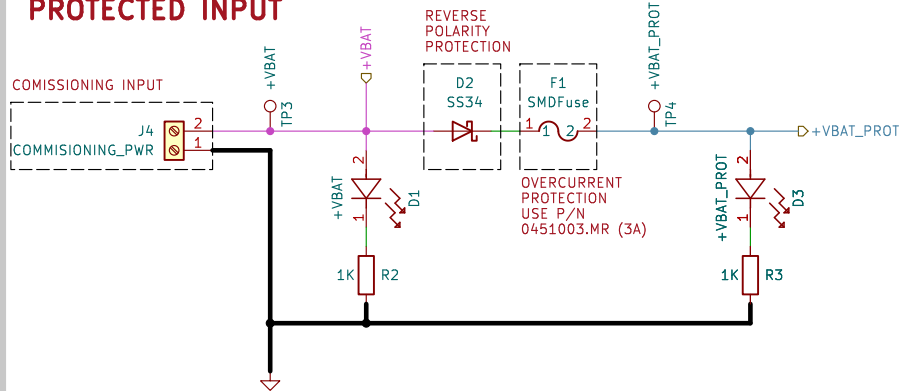
[3] Project Architecture



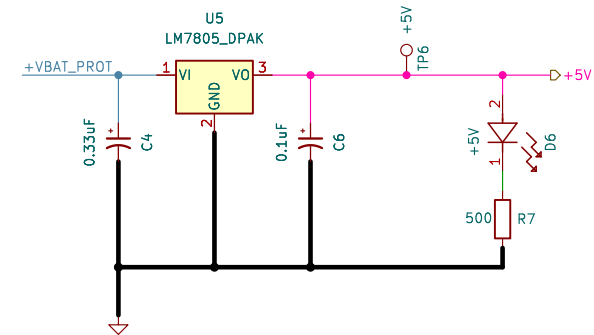
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	TITLE Project Architecture		CREATED DATE 2025-09-12	
	DSGNR. R. HICKS	PART NO	DRAWING NO NU0001-ELEC-ARC-701	
ENG.				
CHKR. Reviewer Name	APPR. Approver Name	KiCad E.D.A. 9.0.3	REVISION 0	SHEET 3 OF 10

[7] Power – Generation

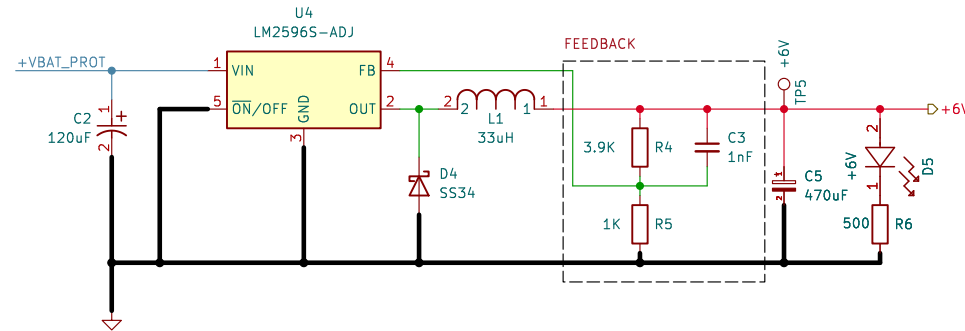
PROTECTED INPUT



+5V GENERATION



+6V GENERATION



NOTES



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TITLE
Power – Generation

CREATED DATE 2025-09-12

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PART NO

DRAWING NO

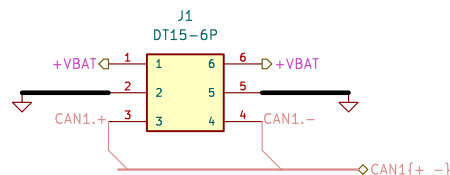
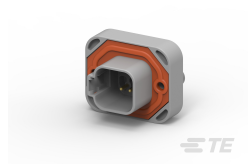
NU0001-ELEC-SCH-701-4(5)

REVISION 0 SHEET 7 OF 10

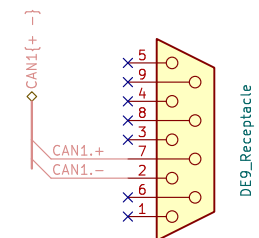
SHEET SIZE

A4

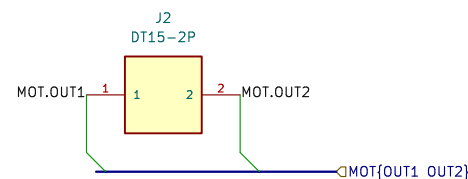
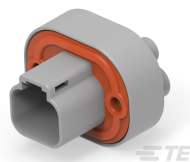
[6] Interface – Connectors



DT15-6P EXPANSION CONNECTOR

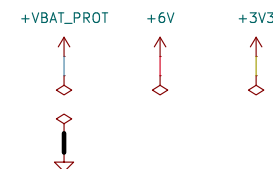


KVASER LEAF DEBUG CONNECTOR



DT15-2P MOTOR CONNECTOR

- H1 MountingHole
- H2 MountingHole
- H3 MountingHole
- H4 MountingHole



- FID1 Fiducial
- FID2 Fiducial
- FID3 Fiducial

NU Teams
Branding Block
Edit in Layout

ERC & MOUNTING HOLES

NOTES



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TITLE
Interface – Connectors

CREATED DATE 2025-09-12

DSGNR. R. HICKS

ENG.

CHKR. Reviewer Name

APPR. Approver Name

PART NO

DRAWING NO

NU0001-ELEC-SCH-701-3(5)

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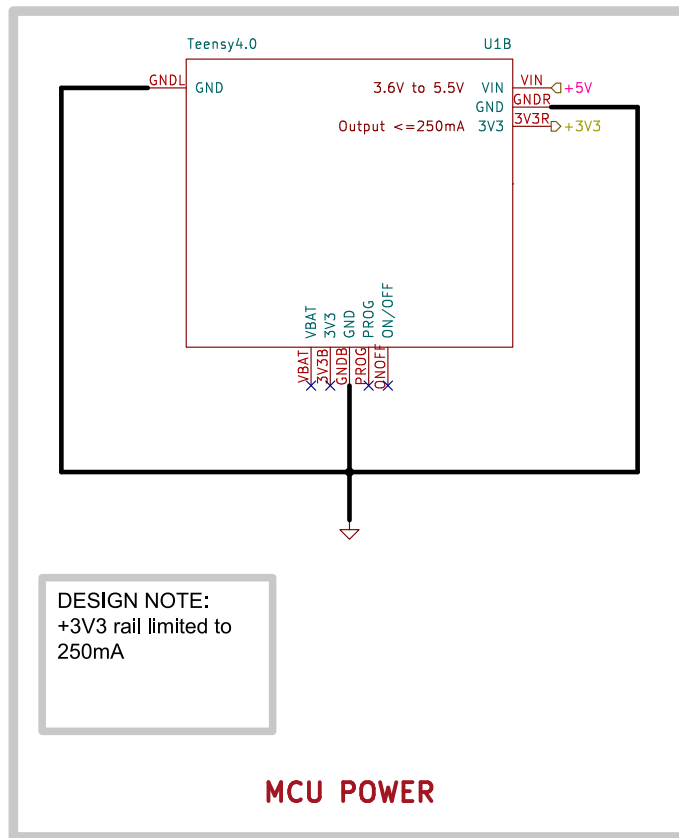
0 SHEET

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SHEET SIZE

A4

[4] MCU – Power

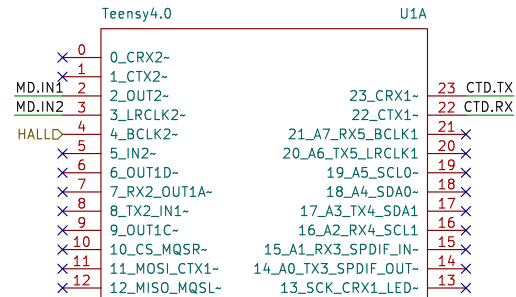


DESIGN NOTE:
+3V3 rail limited to
250mA

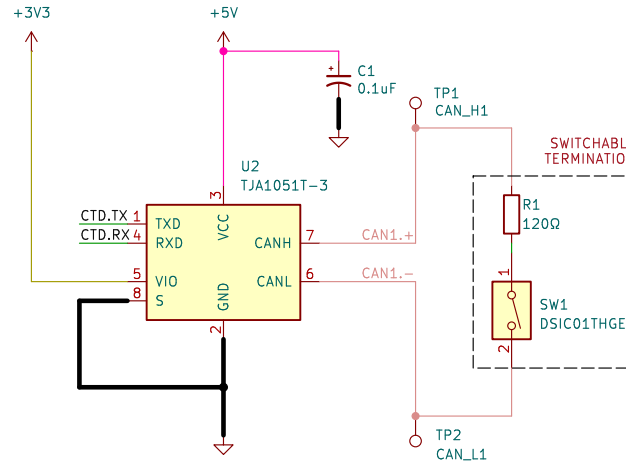
MCU POWER

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	TITLE MCU – Power		CREATED DATE 2025-09-12	
	DSGNR. R. HICKS			
ENG.	PART NO	DRAWING NO		SHEET SIZE
CHKR. Reviewer Name		NU0001-ELEC-SCH-701-1(5)		
APPR. Approver Name	KiCad E.D.A. 9.0.3	REVISION 0	SHEET 4 OF 10	
				A4

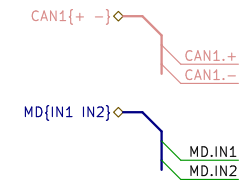
[5] MCU – IO



MCU IO



CAN TRANCEIVER



BUSSES

NOTES



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TITLE
MCU – IO

CREATED DATE 2025-09-12

DSGNR. R. HICKS

ENG.

CHKR. Reviewer Name

APPR. Approver Name

PART NO

DRAWING NO

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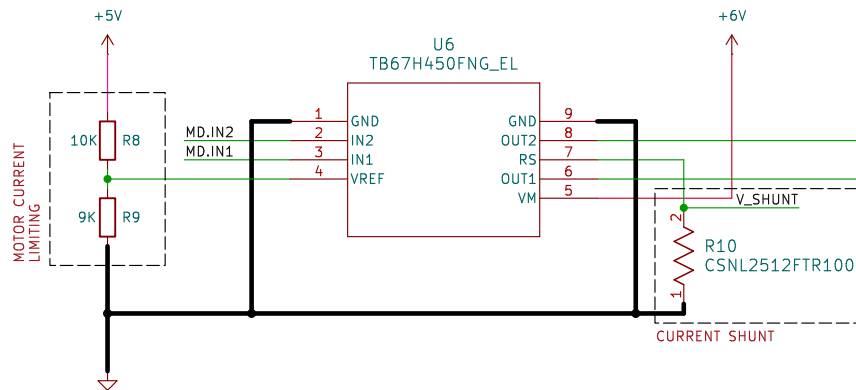
SHEET SIZE

A4

[8] Motor Control

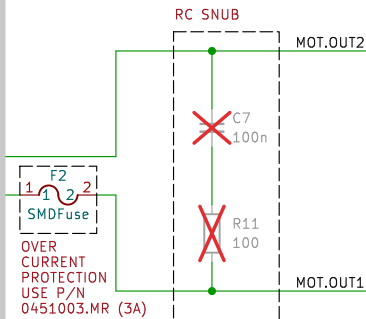
MOTOR SPEED & DIRECTION CONTROL

LAYOUT NOTE:
USE KELVIN FOR
R10



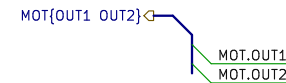
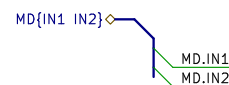
MOTOR

DESIGN NOTE:
FIT RC SNUB IF
RINGING/EMI
OCCURS



MOTOR CURRENT MONITORING

BUSSES



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TITLE
Motor Control

CREATED DATE 2025-09-12

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DRAWING NO

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SHEET SIZE

A4

A

B

C

D

A

B



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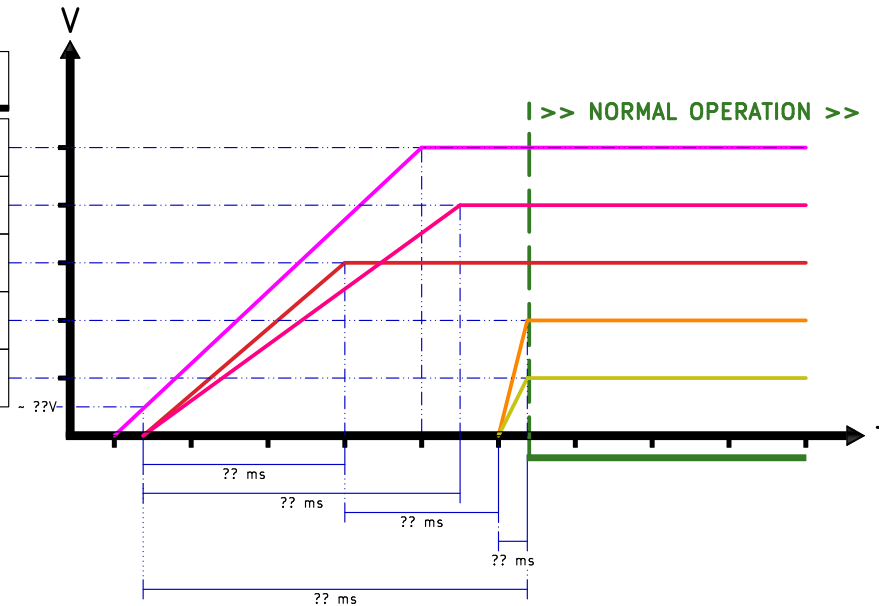
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A4

[9] Power – Sequencing

NAME	SOURCE	LEVEL
+??V	SOURCE NAME	??V ± ??%
+??V	SOURCE NAME	??V ± ??%
+??V	SOURCE NAME	??V ± ??%
+??V	SOURCE NAME	??V ± ??%
+??V	SOURCE NAME	??V ± ??%



NOTES



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TITLE
Power – Sequencing

CREATED DATE 2025-09-12

DSGNR. R. HICKS

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CHKR. Reviewer Name

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DRAWING NO

NU0001-ELEC-SEQ-701

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A4